RiemannFM Mathematical Formalization

RiemannFM: Geometry- and Text-Aware Flow Matching on Product Manifolds with Joint Continuous-Discrete Dynamics for Knowledge Graph Generation — Complete Mathematical Definitions

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1. Base Sets and Indexing Conventions

Definition 1.1 (Base Sets).

- **Entity set**: $\mathcal{V} = \{v_1, v_2, \ldots, v_{|\mathcal{V}|}\}$.
- **Relation type set**: $\mathcal{R} = \{r_1, r_2, \ldots, r_K\}$, where $K \triangleq |\mathcal{R}|$.
- **Knowledge graph**: $\mathcal{K} = (\mathcal{V}, \mathcal{E})$, where the fact set $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{R} \times \mathcal{V}$. A fact $(v_i, r_k, v_j) \in \mathcal{E}$ denotes that entity v_i is directionally connected to entity v_j via relation r_k.

Self-loops: $i = j$ is allowed, i.e., $(v_i, r_k, v_i) \in \mathcal{E}$ is valid.

Multi-relational edges: The same ordered entity pair (v_i, v_j) may be associated with multiple relation types, i.e., $(v_i, r_a, v_j) \in \mathcal{E}$ and $(v_i, r_b, v_j) \in \mathcal{E}$ may coexist with $a \neq b$.

Inverse relations: Semantically inverse relations (e.g., "father" and "child") exist as independent elements in $\mathcal{R}$; no artificial inverse relations are added.

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2. Formal Definition of Subgraphs

Notation: For any positive integer n, let $[n] \triangleq \{1, 2, \ldots, n\}$.

Definition 2.1 (Subgraph). A subgraph $\mathcal{G} = (\mathcal{V}_\mathcal{G}, \mathcal{E}_\mathcal{G})$ is an induced subgraph of $\mathcal{K}$: - $\mathcal{V}_\mathcal{G} \subseteq \mathcal{V}$, $|\mathcal{V}_\mathcal{G}| = N$; - $\mathcal{E}_\mathcal{G} = \{(v_i, r_k, v_j) \in \mathcal{E} \mid v_i, v_j \in \mathcal{V}_\mathcal{G}\}$.

Nodes in $\mathcal{V}_\mathcal{G}$ are re-indexed as $[N]$. Hereafter, $i, j \in [N]$ refer to local indices within the subgraph.

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3. Matrix Representation of Subgraphs

A subgraph $\mathcal{G}$ is encoded as a 5-tuple $(\mathbf{X}, \mathbf{E}, \mathbf{C}_{\mathcal{V}}, \mathbf{C}_{\mathcal{R}}, \mathbf{m})$, corresponding to node manifold coordinates, edge type tensor, node text conditioning matrix, relation text conditioning matrix, and node mask, respectively. Each component is defined in the following subsections.

3.1 Embedding Space: Product Manifold

Definition 3.1 (Lorentz Inner Product). For $\mathbf{a}, \mathbf{b} \in \mathbb{R}^{d_h+1}$:

$$\langle \mathbf{a}, \mathbf{b} \rangle_{\mathrm{L}} = -a_0 b_0 + \sum_{l=1}^{d_h} a_l b_l$$

Definition 3.2 (Product Manifold).

$$\mathcal{M} = \mathbb{H}_{\kappa_h}^{d_h} \times \mathbb{S}_{\kappa_s}^{d_s} \times \mathbb{R}^{d_e}$$

where: - $\mathbb{H}_{\kappa_h}^{d_h} = \{\mathbf{z} \in \mathbb{R}^{d_h+1} \mid \langle \mathbf{z}, \mathbf{z} \rangle_{\mathrm{L}} = 1/\kappa_h,\ z_0 > 0\}$, the d_h-dimensional hyperbolic space (Lorentz model) with curvature $\kappa_h < 0$; - $\mathbb{S}_{\kappa_s}^{d_s} = \{\mathbf{z} \in \mathbb{R}^{d_s+1} \mid \|\mathbf{z}\|_2^2 = 1/\kappa_s\}$, the d_s-dimensional sphere with curvature $\kappa_s > 0$; - $\mathbb{R}^{d_e}$ is the d_e-dimensional Euclidean space.

Total ambient space dimension $D \triangleq (d_h + 1) + (d_s + 1) + d_e$.

Component notation: For $\mathbf{x} \in \mathcal{M}$, denote $\mathbf{x}^{\mathbb{H}} \in \mathbb{R}^{d_h+1}$, $\mathbf{x}^{\mathbb{S}} \in \mathbb{R}^{d_s+1}$, $\mathbf{x}^{\mathbb{R}} \in \mathbb{R}^{d_e}$ as the three subspace components.

3.2 Node Coordinates

Definition 3.3 (Node Coordinates).

$$\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N) \in \mathcal{M}^N$$

where $\mathbf{x}_i = (\mathbf{x}_i^{\mathbb{H}}, \mathbf{x}_i^{\mathbb{S}}, \mathbf{x}_i^{\mathbb{R}}) \in \mathcal{M}$, $i \in [N]$.

3.3 Edge Type Tensor

Definition 3.4 (Edge Type Tensor).

$$\mathbf{E} \in \{0, 1\}^{N \times N \times K}$$

$$\mathbf{E}_{ij}^{(k)} = \mathbb{1}[(v_i, r_k, v_j) \in \mathcal{E}_{\mathcal{G}}]$$

For fixed (i, j), $\mathbf{E}_{ij} \in \{0, 1\}^K$ is a multi-hot vector: $\mathbf{E}_{ij} = \mathbf{0}_K$ indicates no edge from i to j, and $\|\mathbf{E}_{ij}\|_1 \in \{0, 1, \ldots, K\}$ gives the number of relation types from i to j.

Properties: - **Asymmetry**: In general, $\mathbf{E}_{ij} \neq \mathbf{E}_{ji}$; - **Multiplicity**: $\|\mathbf{E}_{ij}\|_1$ may be greater than 1. - **Sparsity**: Average out-degree $\bar{d} = \frac{1}{N} \sum_{i=1}^{N} |\{j \in [N] : \mathbf{E}_{ij} \neq \mathbf{0}_K\}| \ll N$.

3.4 Text Conditioning

Notation: Let Σ be the character alphabet, Σ^* the set of finite strings over Σ, and $\circ$ the string concatenation operator.

Definition 3.5 (Text Conditioning Vector). Let $\phi_{\text{text}} : \Sigma^* \to \mathbb{R}^{d_c}$ be a pretrained text encoder. For each node v_i:

$$\mathbf{c}_{v_i} = \phi_{\text{text}}(\text{label}_{v_i} \circ \text{desc}_{v_i}) \in \mathbb{R}^{d_c}$$

For each relation type r_k:

$$\mathbf{c}_{r_k} = \phi_{\text{text}}(\text{label}_{r_k} \circ \text{desc}_{r_k}) \in \mathbb{R}^{d_c}$$

Hereafter we use the shorthand $\mathbf{c}_i \triangleq \mathbf{c}_{v_i}$.

Definition 3.6 (Text Conditioning Matrices). - Node text matrix: $\mathbf{C}_{\mathcal{V}} = (\mathbf{c}_1, \ldots, \mathbf{c}_N) \in \mathbb{R}^{N \times d_c}$ (subgraph-level); - Relation text matrix: $\mathbf{C}_{\mathcal{R}} = (\mathbf{c}_{r_1}, \ldots, \mathbf{c}_{r_K}) \in \mathbb{R}^{K \times d_c}$ (global, shared across all subgraphs).

3.5 Virtual Node Padding

Definition 3.7 (Virtual Node Padding). Let $N_{\max} \in \mathbb{Z}_{>0}$ be the upper bound on subgraph node count (a preset hyperparameter). For a subgraph with $|\mathcal{V}_{\mathcal{G}}| \leq N_{\max}$, pad $N_{\max} - |\mathcal{V}_{\mathcal{G}}|$ virtual nodes to make the total node count $N_{\max}$:

- Virtual node coordinates: $\mathbf{x}_i = \mathbf{x}_{\varnothing} \in \mathcal{M}$, where the anchor components are

$$\mathbf{x}_{\varnothing}^{\mathbb{H}} = \left(\frac{1}{\sqrt{|\kappa_h|}}, 0, \ldots, 0 \right) \in \mathbb{R}^{d_h+1}, \quad \mathbf{x}_{\varnothing}^{\mathbb{S}} = \left(\frac{1}{\sqrt{\kappa_s}}, 0, \ldots, 0 \right) \in \mathbb{R}^{d_s+1}, \quad \mathbf{x}_{\varnothing}^{\mathbb{R}} = \mathbf{0}_{d_e}$$

- Virtual node edges: $\forall j \in [N_{\max}]$, $\mathbf{E}_{ij} = \mathbf{E}_{ji} = \mathbf{0}_K$;
- Virtual node text: $\mathbf{c}_i = \mathbf{0}_{d_c}$.

One can verify that $\mathbf{x}_{\varnothing}^{\mathbb{H}}$ satisfies $\langle \mathbf{x}_{\varnothing}^{\mathbb{H}}, \mathbf{x}_{\varnothing}^{\mathbb{H}} \rangle_{\text{L}} = -1/|\kappa_h| = 1/\kappa_h$ with a positive first component; and $\mathbf{x}_{\varnothing}^{\mathbb{S}}$ satisfies $\|\mathbf{x}_{\varnothing}^{\mathbb{S}}\|_2^2 = 1/\kappa_s$.

Definition 3.8 (Node Mask).

$$\mathbf{m} \in \{0,1\}^{N_{\max}}, \quad m_i = \begin{cases} 1 & \text{real node} \\ 0 & \text{virtual node} \end{cases}$$

satisfying $\sum_{i=1}^{N_{\max}} m_i = |\mathcal{V}_{\mathcal{G}}|$.

Convention: From this point onward, $N \triangleq N_{\max}$. In all subsequent formulas, $i, j \in [N]$ includes both real and virtual nodes; virtual nodes are identified and masked via $\mathbf{m}$.

3.6 Training Samples and Dataset

Definition 3.9 (Training Sample). A training sample is the 5-tuple encoding of a subgraph:

$$\left(\mathbf{X} \in \mathcal{M}^N,\ \mathbf{E} \in \{0,1\}^{N\times N\times K},\ \mathbf{C}_{\mathcal{V}} \in \mathbb{R}^{N\times d_c},\ \mathbf{C}_{\mathcal{R}} \in \mathbb{R}^{K\times d_c},\ \mathbf{m} \in \{0,1\}^N\right)$$

Definition 3.10 (Training Dataset). $\mathcal{D} = \{\mathcal{G}_1, \mathcal{G}_2, \ldots, \mathcal{G}_{|\mathcal{D}|}\}$ is the collection of subgraphs sampled from the knowledge graph $\mathcal{K}$.

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4. Basic Operations on Manifolds

4.1 Geodesic Distance

Definition 4.1 (Component Geodesic Distances).

- Hyperbolic distance $d_{\mathbb{H}} : \mathbb{H}^{d_h}_{\kappa_h} \times \mathbb{H}^{d_h}_{\kappa_h} \to \mathbb{R}_{\geq 0}$:

$$d_{\mathbb{H}}(\mathbf{a}, \mathbf{b}) = \frac{1}{\sqrt{|\kappa_h|}} \operatorname{arccosh}(\kappa_h \cdot \langle \mathbf{a}, \mathbf{b}\rangle_{\mathrm{L}})$$

  where $\kappa_h \cdot \langle \mathbf{a}, \mathbf{b}\rangle_{\mathrm{L}} \geq 1$ holds for all $\mathbf{a}, \mathbf{b} \in \mathbb{H}^{d_h}_{\kappa_h}$.

- Spherical distance $d_{\mathbb{S}} : \mathbb{S}^{d_s}_{\kappa_s} \times \mathbb{S}^{d_s}_{\kappa_s} \to \mathbb{R}_{\geq 0}$:

$$d_{\mathbb{S}}(\mathbf{a}, \mathbf{b}) = \frac{1}{\sqrt{\kappa_s}} \arccos\big(\kappa_s \cdot \mathbf{a}^\top \mathbf{b}\big)$$

  where $\kappa_s \cdot \mathbf{a}^\top \mathbf{b} \in [-1, 1]$ is guaranteed by the Cauchy-Schwarz inequality.

- Euclidean distance $d_{\mathbb{R}} : \mathbb{R}^{d_e} \times \mathbb{R}^{d_e} \to \mathbb{R}_{\geq 0}$:

$$d_{\mathbb{R}}(\mathbf{a}, \mathbf{b}) = \|\mathbf{a} - \mathbf{b}\|_2$$

Definition 4.2 (Product Manifold Geodesic Distance). $d_{\mathcal{M}} : \mathcal{M} \times \mathcal{M} \to \mathbb{R}_{\geq 0}$:

$$d_{\mathcal{M}}(\mathbf{x}, \mathbf{y}) = \sqrt{d_{\mathbb{H}}(\mathbf{x}^{\mathbb{H}}, \mathbf{y}^{\mathbb{H}})^2 + d_{\mathbb{S}}(\mathbf{x}^{\mathbb{S}}, \mathbf{y}^{\mathbb{S}})^2 + d_{\mathbb{R}}(\mathbf{x}^{\mathbb{R}}, \mathbf{y}^{\mathbb{R}})^2}$$

$d_{\mathcal{M}}$ is a metric on $\mathcal{M}$ (the ℓ_2 combination of component distances preserves the metric axioms).

4.2 Tangent Space

Definition 4.3 (Tangent Space). For $\mathbf{x} \in \mathcal{M}$, the component tangent spaces are: - $T_{\mathbf{x}^{\mathbb{H}}}\mathbb{H} = \{\mathbf{v} \in \mathbb{R}^{d_h+1} \mid \langle \mathbf{v}, \mathbf{x}^{\mathbb{H}}\rangle_{\mathrm{L}} = 0\}$ - $T_{\mathbf{x}^{\mathbb{S}}}\mathbb{S} = \{\mathbf{v} \in \mathbb{R}^{d_s+1} \mid \mathbf{v}^\top \mathbf{x}^{\mathbb{S}} = 0\}$ - $T_{\mathbf{x}^{\mathbb{R}}}\mathbb{R}^{d_e} = \mathbb{R}^{d_e}$

Product tangent space: $T_{\mathbf{x}}\mathcal{M} = T_{\mathbf{x}^{\mathbb{H}}}\mathbb{H} \times T_{\mathbf{x}^{\mathbb{S}}}\mathbb{S} \times \mathbb{R}^{d_e}$. A tangent vector is denoted $\mathbf{v} = (\mathbf{v}^{\mathbb{H}}, \mathbf{v}^{\mathbb{S}}, \mathbf{v}^{\mathbb{R}}) \in T_{\mathbf{x}}\mathcal{M}$.

4.3 Logarithmic Map

The logarithmic map $\log_{\mathbf{x}}(\mathbf{y})$ pulls back a point $\mathbf{y}$ on the manifold to the tangent space at $\mathbf{x}$. It answers the question: "Starting from $\mathbf{x}$, in which direction and how far must one travel to reach $\mathbf{y}$?" The returned tangent vector encodes both direction and distance.

Definition 4.4 (Logarithmic Map). $\log_{\mathbf{x}} : \mathcal{M} \to T_{\mathbf{x}}\mathcal{M}$, with components:

- Hyperbolic component: Let $\mathbf{u} = \mathbf{y}^{\mathbb{H}} - \kappa_h \langle \mathbf{x}^{\mathbb{H}}, \mathbf{y}^{\mathbb{H}} \rangle_{\mathrm{L}} \cdot \mathbf{x}^{\mathbb{H}}$, then

$$\log_{\mathbf{x}^{\mathbb{H}}}(\mathbf{y}^{\mathbb{H}}) = \frac{d_{\mathbb{H}}(\mathbf{x}^{\mathbb{H}}, \mathbf{y}^{\mathbb{H}})}{\|\mathbf{u}\|_{\mathrm{L}}} \cdot \mathbf{u}$$

  where $\|\mathbf{u}\|_{\mathrm{L}} = \sqrt{|\langle \mathbf{u}, \mathbf{u} \rangle_{\mathrm{L}}|}$. One can verify $\langle \mathbf{u}, \mathbf{x}^{\mathbb{H}} \rangle_{\mathrm{L}} = 0$, i.e., $\mathbf{u} \in T_{\mathbf{x}^{\mathbb{H}}}\mathbb{H}$.

- Spherical component: Let $\mathbf{w} = \mathbf{y}^{\mathbb{S}} - \kappa_s (\mathbf{x}^{\mathbb{S}\top} \mathbf{y}^{\mathbb{S}}) \cdot \mathbf{x}^{\mathbb{S}}$, then

$$\log_{\mathbf{x}^{\mathbb{S}}}(\mathbf{y}^{\mathbb{S}}) = \frac{d_{\mathbb{S}}(\mathbf{x}^{\mathbb{S}}, \mathbf{y}^{\mathbb{S}})}{\|\mathbf{w}\|_2} \cdot \mathbf{w}$$

  One can verify $\mathbf{w}^\top \mathbf{x}^{\mathbb{S}} = 0$, i.e., $\mathbf{w} \in T_{\mathbf{x}^{\mathbb{S}}}\mathbb{S}$.

- Euclidean component:

$$\log_{\mathbf{x}^{\mathbb{R}}}(\mathbf{y}^{\mathbb{R}}) = \mathbf{y}^{\mathbb{R}} - \mathbf{x}^{\mathbb{R}}$$

Product: $\log_{\mathbf{x}}(\mathbf{y}) = (\log_{\mathbf{x}^{\mathbb{H}}}(\mathbf{y}^{\mathbb{H}}), \log_{\mathbf{x}^{\mathbb{S}}}(\mathbf{y}^{\mathbb{S}}), \log_{\mathbf{x}^{\mathbb{R}}}(\mathbf{y}^{\mathbb{R}}))$.

Convention: When $\mathbf{y} = \mathbf{x}$, $\log_{\mathbf{x}}(\mathbf{x}) = \mathbf{0} \in T_{\mathbf{x}}\mathcal{M}$.

4.4 Exponential Map

The exponential map $\exp_{\mathbf{x}}(\mathbf{v})$ pushes a tangent vector $\mathbf{v}$ back onto the manifold. It answers the question: "Starting from $\mathbf{x}$, following the direction of tangent vector $\mathbf{v}$ for a geodesic distance of $\|\mathbf{v}\|$, which point on the manifold is reached?"

Definition 4.5 (Exponential Map). $\exp_{\mathbf{x}} : T_{\mathbf{x}}\mathcal{M} \to \mathcal{M}$, with components:

- Hyperbolic component:

$$\exp_{\mathbf{x}^{\mathbb{H}}}(\mathbf{v}^{\mathbb{H}}) = \cosh(\sqrt{|\kappa_h|}\, \|\mathbf{v}^{\mathbb{H}}\|_{\mathrm{L}})\, \mathbf{x}^{\mathbb{H}} + \frac{\sinh(\sqrt{|\kappa_h|}\, \|\mathbf{v}^{\mathbb{H}}\|_{\mathrm{L}})}{\sqrt{|\kappa_h|}\, \|\mathbf{v}^{\mathbb{H}}\|_{\mathrm{L}}}\, \mathbf{v}^{\mathbb{H}}$$

- Spherical component:

$$\exp_{\mathbf{x}^{\mathbb{S}}}(\mathbf{v}^{\mathbb{S}}) = \cos(\sqrt{\kappa_s}\, \|\mathbf{v}^{\mathbb{S}}\|_2)\, \mathbf{x}^{\mathbb{S}} + \frac{\sin(\sqrt{\kappa_s}\, \|\mathbf{v}^{\mathbb{S}}\|_2)}{\sqrt{\kappa_s}\, \|\mathbf{v}^{\mathbb{S}}\|_2}\, \mathbf{v}^{\mathbb{S}}$$

- Euclidean component:

$$\exp_{\mathbf{x}^{\mathbb{R}}}(\mathbf{v}^{\mathbb{R}}) = \mathbf{x}^{\mathbb{R}} + \mathbf{v}^{\mathbb{R}}$$

Product: $\exp_{\mathbf{x}}(\mathbf{v}) = (\exp_{\mathbf{x}^{\mathbb{H}}}(\mathbf{v}^{\mathbb{H}}), \exp_{\mathbf{x}^{\mathbb{S}}}(\mathbf{v}^{\mathbb{S}}), \exp_{\mathbf{x}^{\mathbb{R}}}(\mathbf{v}^{\mathbb{R}}))$.

Convention: When $\mathbf{v} = \mathbf{0}$, $\exp_{\mathbf{x}}(\mathbf{0}) = \mathbf{x}$ (in each component, $\sinh(0)/0$ and $\sin(0)/0$ are taken as the limit value 1).

Proposition 4.1 (Inverse Property). Within the injectivity radius (hyperbolic and Euclidean: ∞; spherical: $\pi/\sqrt{\kappa_s}$), $\exp_{\mathbf{x}} \circ \log_{\mathbf{x}} = \mathrm{id}_{\mathcal{M}}$ and $\log_{\mathbf{x}} \circ \exp_{\mathbf{x}} = \mathrm{id}_{T_{\mathbf{x}}\mathcal{M}}$.

4.5 Tangent Space Riemannian Norm

Definition 4.6 (Tangent Space Riemannian Norm). For $\mathbf{v} = (\mathbf{v}^{\mathbb{H}}, \mathbf{v}^{\mathbb{S}}, \mathbf{v}^{\mathbb{R}}) \in T_{\mathbf{x}}\mathcal{M}$:

$$\|\mathbf{v}\|_{T_{\mathbf{x}}\mathcal{M}} = \sqrt{\|\mathbf{v}^{\mathbb{H}}\|_{\mathrm{L}}^2 + \|\mathbf{v}^{\mathbb{S}}\|_2^2 + \|\mathbf{v}^{\mathbb{R}}\|_2^2}$$

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5. Model Architecture: RieFormer

5.1 Overall Definition

Definition 5.1 (RieFormer). RieFormer is a mapping with parameters θ:

$$f_\theta(\mathbf{X}_t, \mathbf{E}_t, t, \mathbf{C}_{\mathcal{V}}, \mathbf{C}_{\mathcal{R}}, \mathbf{m}) = (\hat{\mathbf{V}}, \hat{\mathbf{P}})$$

Inputs: $\mathbf{X}_t \in \mathcal{M}^N$, $\mathbf{E}_t \in \{0,1\}^{N\times N\times K}$, $t \in [0,1]$, $\mathbf{C}_{\mathcal{V}} \in \mathbb{R}^{N\times d_c}$, $\mathbf{C}_{\mathcal{R}} \in \mathbb{R}^{K\times d_c}$, $\mathbf{m} \in \{0,1\}^N$.

Outputs: - $\hat{\mathbf{v}}_i \in T_{\mathbf{x}_{t,i}}\mathcal{M}$, the predicted tangent vector for node i, $i \in [N]$; - $\hat{\mathbf{P}} \in [0,1]^{N\times N\times K}$, where $\hat{\mathbf{P}}_{ij}^{(k)}$ is the predicted edge type probability.

Let $\mathbf{X}_t = (\mathbf{x}_{t,1}, \ldots, \mathbf{x}_{t,N}) \in \mathcal{M}^N$, where $\mathbf{x}_{t,i} \in \mathcal{M}$ is the manifold coordinate of node i at time t, obtained via geodesic interpolation from noise $\mathbf{x}_{0,i}$ to data $\mathbf{x}_{1,i}$.

θ includes all learnable parameters, including curvatures κ_h, κ_s.

5.2 Input Encoding Layer

Notation: $[\mathbf{a}\|\mathbf{b}]$ denotes vector concatenation.

Definition 5.2 (Coordinate Projection). $\pi : \mathcal{M} \to \mathbb{R}^D$:

$$\pi(\mathbf{x}) = [\mathbf{x}^{\mathbb{H}}\|\mathbf{x}^{\mathbb{S}}\|\mathbf{x}^{\mathbb{R}}]$$

Definition 5.3 (Node Initial Embedding).

$$\mathbf{h}_i^{V,(0)} = \mathrm{MLP}_{\mathrm{node}}([\pi(\mathbf{x}_{t,i})\|\mathbf{c}_i\|m_i]) \in \mathbb{R}^{d_v}$$

where the input dimension is $D + d_c + 1$, and d_v is the node hidden dimension.

Definition 5.4 (Edge Initial Embedding).

$$\mathbf{h}_{ij}^{E,(0)} = \mathrm{MLP}_{\mathrm{edge}}([\mathbf{E}_{t,ij}\mathbf{W}_{\mathrm{rel}} \| \mathbf{E}_{t,ij}\mathbf{C}_{\mathcal{R}}]) \in \mathbb{R}^{d_{e'}}$$

where $\mathbf{W}_{\mathrm{rel}} \in \mathbb{R}^{K \times d_r}$ is a learnable relation embedding matrix, $\mathbf{E}_{t,ij}\mathbf{W}_{\mathrm{rel}} \in \mathbb{R}^{d_r}$ is the sum of active relation type embeddings, and $\mathbf{E}_{t,ij}\mathbf{C}_{\mathcal{R}} \in \mathbb{R}^{d_c}$ is the sum of active relation type text embeddings. Input dimension is $d_r + d_c$.

Definition 5.5 (Time Embedding). Let $d_t \in \mathbb{Z}_{>0}$ be even; sinusoidal positional encoding:

$$\boldsymbol{\psi}(t) = [\sin(\omega_1 t), \cos(\omega_1 t), \ldots, \sin(\omega_{d_t/2} t), \cos(\omega_{d_t/2} t)] \in \mathbb{R}^{d_t}$$

where $\omega_l = 10000^{-2l/d_t}$, $l \in [d_t/2]$. Linear projection:

$$\mathbf{t}_{\mathrm{emb}} = \mathbf{W}_t \boldsymbol{\psi}(t) + \mathbf{b}_t \in \mathbb{R}^{d_v}$$

where $\mathbf{W}_t \in \mathbb{R}^{d_v \times d_t}$, $\mathbf{b}_t \in \mathbb{R}^{d_v}$.

5.3 RieFormer Block

RieFormer consists of $L \in \mathbb{Z}_{>0}$ identically structured blocks stacked together. Layer l ($l \in [L]$) receives node embeddings $\mathbf{H}^{V,(l-1)} \in \mathbb{R}^{N \times d_v}$ and edge embeddings $\mathbf{H}^{E,(l-1)} \in \mathbb{R}^{N \times N \times d_{e'}}$, and outputs $\mathbf{H}^{V,(l)} \in \mathbb{R}^{N \times d_v}$ and $\mathbf{H}^{E,(l)} \in \mathbb{R}^{N \times N \times d_{e'}}$.

Each block executes five submodules in the following order:

- A. Manifold-Aware Multi-Head Attention (Manifold RoPE + Geodesic Kernel) → updates node embeddings
- B. ATH-Norm → normalizes node embeddings
- C. Edge Stream Self-Update → updates edge embeddings
- D. Bidirectional Cross-Interaction → mutual injection between node and edge embeddings
- E. Text Conditioning Injection → injects text semantics via cross-attention

5.3.1 Submodule A: Manifold-Aware Multi-Head Attention

5.3.1 Submodule A: Manifold-Aware Multi-Head Attention Let $n_h \in \mathbb{Z}_{>0}$ be the number of attention heads, $d_{\mathrm{head}} = d_v/n_h$ (requiring $n_h \mid d_v$ and $2 \mid d_{\mathrm{head}}$).

Definition 5.6 (Manifold RoPE). For the s-th attention head ($s \in [n_h]$), frequencies $\omega_l^{(s)} = 10000^{-2l/d_{\mathrm{head}}}$, $l \in [d_{\mathrm{head}}/2]$. For a node pair (i, j), define the angle:

$$\theta_{ij,l}^{(s)} = \omega_l^{(s)} \cdot d_{\mathcal{M}}(\mathbf{x}_{t,i}, \mathbf{x}_{t,j})$$

The rotation matrix $\mathbf{R}(\boldsymbol{\theta}_{ij}^{(s)}) \in \mathbb{R}^{d_{\text{head}} \times d_{\text{head}}}$ is a block-diagonal matrix composed of $d_{\text{head}}/2$ rotation blocks of size 2×2:

$$\mathbf{R}(\boldsymbol{\theta}_{ij}^{(s)}) = \operatorname{diag}\left(\begin{pmatrix} \cos\theta_{ij,1}^{(s)} & -\sin\theta_{ij,1}^{(s)} \\ \sin\theta_{ij,1}^{(s)} & \cos\theta_{ij,1}^{(s)} \end{pmatrix}, \ldots, \begin{pmatrix} \cos\theta_{ij,d_{\text{head}}/2}^{(s)} & -\sin\theta_{ij,d_{\text{head}}/2}^{(s)} \\ \sin\theta_{ij,d_{\text{head}}/2}^{(s)} & \cos\theta_{ij,d_{\text{head}}/2}^{(s)} \end{pmatrix} \right)$$

Definition 5.7 (Geodesic Kernel). For the s-th attention head:

$$\kappa^{(s)}(\mathbf{x}_{t,i}, \mathbf{x}_{t,j}) = w_{\mathbb{H}}^{(s)} \kappa_{\mathbb{H}}(\mathbf{x}_{t,i}^{\mathbb{H}}, \mathbf{x}_{t,j}^{\mathbb{H}}) + w_{\mathbb{S}}^{(s)} \kappa_{\mathbb{S}}(\mathbf{x}_{t,i}^{\mathbb{S}}, \mathbf{x}_{t,j}^{\mathbb{S}}) + w_{\mathbb{R}}^{(s)} \kappa_{\mathbb{R}}(\mathbf{x}_{t,i}^{\mathbb{R}}, \mathbf{x}_{t,j}^{\mathbb{R}})$$

where: - $\kappa_{\mathbb{H}}(\mathbf{a}, \mathbf{b}) = -d_{\mathbb{H}}(\mathbf{a}, \mathbf{b})$ - $\kappa_{\mathbb{S}}(\mathbf{a}, \mathbf{b}) = \kappa_s \cdot \mathbf{a}^\top \mathbf{b}$ - $\kappa_{\mathbb{R}}(\mathbf{a}, \mathbf{b}) = -\|\mathbf{a} - \mathbf{b}\|_2^2$

$w_{\mathbb{H}}^{(s)}, w_{\mathbb{S}}^{(s)}, w_{\mathbb{R}}^{(s)} \in \mathbb{R}$ are per-head learnable weights.

Definition 5.8 (Manifold-Aware Attention). Query/key/value projections ($s \in [n_h]$):

$$\mathbf{q}_i^{(s)} = \mathbf{W}_Q^{(s)} \mathbf{h}_i^{V,(l-1)}, \quad \mathbf{k}_j^{(s)} = \mathbf{W}_K^{(s)} \mathbf{h}_j^{V,(l-1)}, \quad \mathbf{v}_j^{(s)} = \mathbf{W}_V^{(s)} \mathbf{h}_j^{V,(l-1)} \in \mathbb{R}^{d_{\text{head}}}$$

where $\mathbf{W}_Q^{(s)}, \mathbf{W}_K^{(s)}, \mathbf{W}_V^{(s)} \in \mathbb{R}^{d_{\text{head}} \times d_v}$.

Attention score:

$$a_{ij}^{(s)} = \frac{(\mathbf{R}(\boldsymbol{\theta}_{ij}^{(s)}) \mathbf{q}_i^{(s)})^\top \mathbf{k}_j^{(s)}}{\sqrt{d_{\text{head}}}} + \beta^{(s)} \kappa^{(s)}(\mathbf{x}_{t,i}, \mathbf{x}_{t,j}) + \mathbf{w}_b^{(s)\top} \mathbf{h}_{ij}^{E,(l-1)}$$

where $\beta^{(s)} \in \mathbb{R}$ is a learnable scaling coefficient, and $\mathbf{w}_b^{(s)} \in \mathbb{R}^{d_{e'}}$ is the edge bias weight.

Attention weights: $\alpha_{ij}^{(s)} = \operatorname{softmax}_j(a_{ij}^{(s)})$.

Euclidean aggregation:

$$\mathbf{o}_i^{(s)} = \sum_{j=1}^{N} \alpha_{ij}^{(s)} \cdot \mathbf{v}_j^{(s)} \in \mathbb{R}^{d_{\text{head}}}$$

Multi-head concatenation and projection:

$$\text{MHA}_i = \mathbf{W}_O [\mathbf{o}_i^{(1)} \| \cdots \| \mathbf{o}_i^{(n_h)}] \in \mathbb{R}^{d_v}$$

where $\mathbf{W}_O \in \mathbb{R}^{d_v \times d_v}$.

Residual connection: $\tilde{\mathbf{h}}_i^V = \mathbf{h}_i^{V,(l-1)} + \text{MHA}_i$.

5.3.3 Submodule C: Edge Stream Self-Update Edge embeddings are independently updated via factorized attention: for edge (i, j), information is aggregated separately from the other outgoing edges of head node i and the other incoming edges of tail node j.

Definition 5.11 (Factorized Edge Attention). For edge (i, j):

Head-side aggregation (other outgoing edges of i):

$$\mathbf{g}_{ij}^{\text{head}} = \sum_{p \in [N] \setminus \{j\}} \gamma_{ip \to ij}^{\text{head}} \cdot \mathbf{W}_{\text{Ev}}^{\text{head}} \mathbf{h}_{ip}^{E,(l-1)}$$

Tail-side aggregation (other incoming edges of j):

$$\mathbf{g}_{ij}^{\text{tail}} = \sum_{p \in [N] \setminus \{i\}} \gamma_{pj \to ij}^{\text{tail}} \cdot \mathbf{W}_{\text{Ev}}^{\text{tail}} \mathbf{h}_{pj}^{E,(l-1)}$$

Attention weights:

$$\gamma_{ip \to ij}^{\text{head}} = \text{softmax}_p \left(\frac{(\mathbf{W}_{\text{Eq}}^{\text{head}} \mathbf{h}_{ij}^{E,(l-1)})^\top (\mathbf{W}_{\text{Ek}}^{\text{head}} \mathbf{h}_{ip}^{E,(l-1)})}{\sqrt{d_{e'}}} \right)$$

The tail-side $\gamma_{pj \to ij}^{\text{tail}}$ is defined analogously, using independent weight matrices $\mathbf{W}_{\text{Eq}}^{\text{tail}}, \mathbf{W}_{\text{Ek}}^{\text{tail}}$.

All weight matrices $\mathbf{W}_{\text{Eq}}^{\text{head}}, \mathbf{W}_{\text{Ek}}^{\text{head}}, \mathbf{W}_{\text{Ev}}^{\text{head}}, \mathbf{W}_{\text{Eq}}^{\text{tail}}, \mathbf{W}_{\text{Ek}}^{\text{tail}}, \mathbf{W}_{\text{Ev}}^{\text{tail}} \in \mathbb{R}^{d_{e'} \times d_{e'}}$.

Residual update:

$$\tilde{\mathbf{h}}_{ij}^{E} = \mathbf{h}_{ij}^{E,(l-1)} + \text{MLP}_{\text{E-up}} \left([\mathbf{h}_{ij}^{E,(l-1)} \| \mathbf{g}_{ij}^{\text{head}} \| \mathbf{g}_{ij}^{\text{tail}}] \right)$$

where $\text{MLP}_{\text{E-up}} : \mathbb{R}^{3d_{e'}} \to \mathbb{R}^{d_{e'}}$.

Followed by LayerNorm: $\bar{\mathbf{h}}_{ij}^{E} = \text{LN}(\tilde{\mathbf{h}}_{ij}^{E})$.

5.3.4 Submodule D: Bidirectional Cross-Interaction **Definition 5.12 (Edge→Node Aggregation).** For node i, aggregate its associated edge embeddings:

$$\hat{\mathbf{h}}_i^V = \bar{\mathbf{h}}_i^V + \text{MLP}_{E \to V} \left(\sum_{j=1}^{N} \alpha_{ij}^{E \to V} \cdot \mathbf{W}_{\text{Ev2n}} \bar{\mathbf{h}}_{ij}^{E} \right)$$

where $\mathbf{W}_{\text{Ev2n}} \in \mathbb{R}^{d_v \times d_{e'}}$, $\text{MLP}_{E \to V} : \mathbb{R}^{d_v} \to \mathbb{R}^{d_v}$. Attention weights:

$$\alpha_{ij}^{E \to V} = \text{softmax}_j \left(\frac{(\mathbf{W}_Q^{E \to V} \bar{\mathbf{h}}_i^V)^\top (\mathbf{W}_K^{E \to V} \bar{\mathbf{h}}_{ij}^{E})}{\sqrt{d_v}} \right)$$

where $\mathbf{W}_Q^{E\to V} \in \mathbb{R}^{d_v \times d_v}$, $\mathbf{W}_K^{E\to V} \in \mathbb{R}^{d_v \times d_{e'}}$.

Definition 5.13 (Node→Edge Injection). For edge (i, j):

$$\hat{\mathbf{h}}_{ij}^E = \bar{\mathbf{h}}_{ij}^E + \mathrm{MLP}_{V\to E}\Big([\hat{\mathbf{h}}_i^V \| \hat{\mathbf{h}}_j^V \| \hat{\mathbf{h}}_i^V \odot \hat{\mathbf{h}}_j^V]\Big)$$

where $\mathrm{MLP}_{V\to E} : \mathbb{R}^{3d_v} \to \mathbb{R}^{d_{e'}}$.

5.3.5 Submodule E: Text Conditioning Injection **Definition 5.14 (Text Cross-Attention).** For node i, using the node embedding as query and text conditioning as key/value:

$$\mathbf{q}_i^{\text{text}} = \mathbf{W}_Q^{\text{text}} \hat{\mathbf{h}}_i^V, \quad \mathbf{k}_j^{\text{text}} = \mathbf{W}_K^{\text{text}} \mathbf{c}_j, \quad \mathbf{v}_j^{\text{text}} = \mathbf{W}_V^{\text{text}} \mathbf{c}_j$$

where $\mathbf{W}_Q^{\text{text}} \in \mathbb{R}^{d_v \times d_v}$, $\mathbf{W}_K^{\text{text}}, \mathbf{W}_V^{\text{text}} \in \mathbb{R}^{d_v \times d_c}$.

Cross-attention:

$$\mathrm{CrossAttn}_i = \sum_{j=1}^{N} \mathrm{softmax}_j \left(\frac{\mathbf{q}_i^{\text{text}\top} \mathbf{k}_j^{\text{text}}}{\sqrt{d_v}} \right) \cdot \mathbf{v}_j^{\text{text}}$$

Definition 5.15 (Layer l Output). - Node: $\mathbf{h}_i^{V,(l)} = \mathrm{LN}(\hat{\mathbf{h}}_i^V + \mathrm{CrossAttn}_i)$ - Edge: $\mathbf{h}_{ij}^{E,(l)} = \hat{\mathbf{h}}_{ij}^E$

5.4 Output Projection Layer

5.4.1 Vector Field Output Head **Definition 5.16 (Tangent Space Projection).** For any $\hat{\mathbf{u}} \in \mathbb{R}^D$, split it in reverse order of the coordinate projection π into $\hat{\mathbf{u}}^{\mathbb{H}} \in \mathbb{R}^{d_h+1}$, $\hat{\mathbf{u}}^{\mathbb{S}} \in \mathbb{R}^{d_s+1}$, $\hat{\mathbf{u}}^{\mathbb{R}} \in \mathbb{R}^{d_e}$. Project onto $T_{\mathbf{x}_t}\mathcal{M}$:

$$\hat{\mathbf{v}}^{\mathbb{H}} = \hat{\mathbf{u}}^{\mathbb{H}} - \kappa_h \langle \hat{\mathbf{u}}^{\mathbb{H}}, \mathbf{x}_t^{\mathbb{H}} \rangle_{\mathrm{L}} \cdot \mathbf{x}_t^{\mathbb{H}} \in T_{\mathbf{x}_t^{\mathbb{H}}}\mathbb{H}$$

$$\hat{\mathbf{v}}^{\mathbb{S}} = \hat{\mathbf{u}}^{\mathbb{S}} - \kappa_s (\mathbf{x}_t^{\mathbb{S}\top} \hat{\mathbf{u}}^{\mathbb{S}}) \cdot \mathbf{x}_t^{\mathbb{S}} \in T_{\mathbf{x}_t^{\mathbb{S}}}\mathbb{S}$$

$$\hat{\mathbf{v}}^{\mathbb{R}} = \hat{\mathbf{u}}^{\mathbb{R}} \in \mathbb{R}^{d_e}$$

Definition 5.17 (Vector Field Prediction). For node i:

$$\hat{\mathbf{u}}_i = \mathrm{MLP}_{\text{vec}}(\mathbf{h}_i^{V,(L)}) \in \mathbb{R}^D$$

where $\mathrm{MLP}_{\text{vec}} : \mathbb{R}^{d_v} \to \mathbb{R}^D$. After projection via Definition 5.16, we obtain $\hat{\mathbf{v}}_i = (\hat{\mathbf{v}}_i^{\mathbb{H}}, \hat{\mathbf{v}}_i^{\mathbb{S}}, \hat{\mathbf{v}}_i^{\mathbb{R}}) \in T_{\mathbf{x}_{t,i}}\mathcal{M}$.

5.4.2 Edge Type Output Head **Definition 5.18 (Relation Interaction Layer).** For edge (i, j) and each relation type $k \in [K]$, construct the candidate relation feature:

$$\mathbf{r}_{ij}^{(k)} = \mathrm{MLP}_{\text{rel-proj}}\Big([\mathbf{h}_{ij}^{E,(L)} \| \mathbf{c}_{r_k}]\Big) \in \mathbb{R}^{d_{e'}}$$

where $\mathrm{MLP}_{\text{rel-proj}} : \mathbb{R}^{d_{e'}+d_c} \to \mathbb{R}^{d_{e'}}$.

The K candidate features $\{\mathbf{r}_{ij}^{(1)}, \ldots, \mathbf{r}_{ij}^{(K)}\}$ interact via a relation Transformer (self-attention over the relation dimension) to produce $\tilde{\mathbf{r}}_{ij}^{(k)} \in \mathbb{R}^{d_{e'}}$, $k \in [K]$.

Definition 5.19 (Edge Type Probability).

$$\hat{\mathbf{P}}_{ij}^{(k)} = \sigma(\mathbf{w}_{\text{cls}}^{\top} \tilde{\mathbf{r}}_{ij}^{(k)} + b_{\text{cls}}) \in [0, 1]$$

where $\sigma(\cdot)$ is the sigmoid function, $\mathbf{w}_{\text{cls}} \in \mathbb{R}^{d_{e'}}$, $b_{\text{cls}} \in \mathbb{R}$. Each relation type is predicted independently (corresponding to multi-hot labels).

5.5 Permutation Equivariance

Proposition 5.1 (Permutation Equivariance). For any permutation matrix $\mathbf{\Pi} \in \{0, 1\}^{N \times N}$:

$$f_\theta(\mathbf{\Pi}\mathbf{X}_t, \mathbf{\Pi}\mathbf{E}_t\mathbf{\Pi}^{\top}, t, \mathbf{\Pi}\mathbf{C}_{\mathcal{V}}, \mathbf{C}_{\mathcal{R}}, \mathbf{\Pi}\mathbf{m}) = (\mathbf{\Pi}\hat{\mathbf{V}}, \mathbf{\Pi}\hat{\mathbf{P}}\mathbf{\Pi}^{\top})$$

where $\mathbf{C}_{\mathcal{R}}$ and t are unaffected by the permutation.

---

6. Joint Discrete-Continuous Flow Matching

Convention: $t = 0$ corresponds to the noise end, $t = 1$ corresponds to the data end. $\mathbf{X}_1, \mathbf{E}_1$ are training data (from subgraphs), and $\mathbf{X}_0, \mathbf{E}_0$ are noise samples.

6.1 Noise Prior Distribution

Definition 6.1 (Continuous Noise Prior). For each node $i \in [N]$, each component is sampled independently: - Hyperbolic component: $\mathbf{x}_{0,i}^{\mathbb{H}} \sim \mathrm{Uniform}(B_{\mathbb{H}}(\mathbf{x}_{\varnothing}^{\mathbb{H}}, R_{\mathbb{H}}))$, i.e., uniformly sampled with respect to the Riemannian volume measure within a geodesic ball centered at the anchor $\mathbf{x}_{\varnothing}^{\mathbb{H}}$ with geodesic radius $R_{\mathbb{H}} > 0$; - Spherical component: $\mathbf{x}_{0,i}^{\mathbb{S}} \sim \mathrm{Uniform}(\mathbb{S}_{\kappa_s}^{d_s})$, i.e., the uniform distribution with respect to the spherical volume measure; - Euclidean component: $\mathbf{x}_{0,i}^{\mathbb{R}} \sim \mathcal{N}(\mathbf{0}, \sigma_0^2 \mathbf{I}_{d_e})$.

where $R_{\mathbb{H}} > 0$ and $\sigma_0 > 0$ are hyperparameters. The continuous noise prior is denoted $p_0^{\mathcal{M}}$.

Definition 6.2 (Discrete Noise Prior). For each pair $(i, j) \in [N]^2$ and each relation type $k \in [K]$, independently sample:

$$\mathbf{E}_{0,ij}^{(k)} \sim \text{Bernoulli}(\rho_k)$$

where the marginal frequency is:

$$\rho_k = \frac{\sum_{\mathcal{G} \in \mathcal{D}} \sum_{i=1}^{N} \sum_{j=1}^{N} \mathbf{E}_{1,ij}^{(k)}}{|\mathcal{D}| \cdot N^2}$$

i.e., the average occurrence frequency of relation type r_k in the training set (virtual node edges are zero and do not affect the numerator).

6.2 Forward Interpolation Process

Definition 6.3 (Continuous Interpolation). For each node $i \in [N]$, interpolate along the geodesic between $\mathbf{x}_{0,i}$ (noise) and $\mathbf{x}_{1,i}$ (data):

$$\mathbf{x}_{t,i} = \exp_{\mathbf{x}_{0,i}}\Big(t \cdot \log_{\mathbf{x}_{0,i}}(\mathbf{x}_{1,i})\Big) \in \mathcal{M}, \quad t \in [0, 1]$$

Boundary conditions: $\mathbf{x}_{0,i} = \mathbf{x}_{0,i}$ (noise), $\mathbf{x}_{1,i} = \mathbf{x}_{1,i}$ (data).

Definition 6.4 (Discrete Interpolation). For each pair $(i, j) \in [N]^2$, independently sample a mask:

$$z_{ij} \sim \text{Bernoulli}(t)$$

$$\mathbf{E}_{t,ij} = z_{ij} \cdot \mathbf{E}_{1,ij} + (1 - z_{ij}) \cdot \mathbf{E}_{0,ij}$$

All K relation types for the same pair (i, j) share the same z_{ij}, i.e., they are taken entirely from the data end or the noise end. Different directions z_{ij} and z_{ji} are sampled independently.

6.3 Conditional Targets

Definition 6.5 (Conditional Vector Field Target). For each node $i \in [N]$:

$$\mathbf{u}_{t,i} = \frac{1}{1-t} \log_{\mathbf{x}_{t,i}}(\mathbf{x}_{1,i}) \in T_{\mathbf{x}_{t,i}}\mathcal{M}$$

This is the tangent vector pointing from the current position $\mathbf{x}_{t,i}$ toward the data point $\mathbf{x}_{1,i}$, scaled by the remaining time $1 - t$ to form a velocity: following this velocity for time $1-t$ exactly reaches $\mathbf{x}_{1,i}$. During training, $t \sim \text{Uniform}(0, 1-\epsilon_t)$, where $\epsilon_t > 0$ is a small constant to avoid the singularity as $t \to 1$.

Definition 6.6 (Edge Type Target). For each pair $(i, j) \in [N]^2$ and each relation type $k \in [K]$:

$$\mathbf{E}_{1,ij}^{(k)} \in \{0, 1\}$$

i.e., the ground-truth edge type label from the data end.

6.4 Training Loss Function

Definition 6.7 (Manifold Vector Field Loss).

$$\mathcal{L}_{\text{cont}} = \frac{1}{N} \sum_{i=1}^{N} m_i \cdot \|\hat{\mathbf{v}}_i - \mathbf{u}_{t,i}\|^2_{T_{\mathbf{x}_{t,i}}\mathcal{M}}$$

where $\|\cdot\|_{T_{\mathbf{x}_{t,i}}\mathcal{M}}$ is the tangent space Riemannian norm from Definition 4.6. Virtual nodes ($m_i = 0$) do not contribute to the loss.

Definition 6.8 (Edge Type Loss). Let $\mathcal{S} \subseteq [N]^2$ be the sampled set of edge pairs.

$$\mathcal{L}_{\text{disc}} = \frac{1}{|\mathcal{S}|} \sum_{(i,j)\in\mathcal{S}} \sum_{k=1}^{K} \text{BCE}_{\text{w}}(\hat{\mathbf{P}}_{ij}^{(k)}, \mathbf{E}_{1,ij}^{(k)})$$

where the weighted binary cross-entropy is:

$$\text{BCE}_{\text{w}}(\hat{p}, y) = -\left[w_k^+ \cdot y \cdot \log(\hat{p} + \epsilon_p) + (1-y) \cdot \log(1 - \hat{p} + \epsilon_p)\right]$$

The positive sample weight $w_k^+ = \min\left(\frac{1-\rho_k}{\rho_k},\, w_{\max}\right)$ corrects for class imbalance, $w_{\max} > 0$ is the truncation upper bound, and $\epsilon_p > 0$ is a numerical stability constant.

Definition 6.9 (Graph-Text Contrastive Loss). Project data-end coordinates into the alignment space:

$$\mathbf{g}_i = \text{MLP}_{\text{proj}}(\pi(\mathbf{x}_{1,i})) \in \mathbb{R}^{d_{\text{a}}}$$

where $\text{MLP}_{\text{proj}} : \mathbb{R}^D \to \mathbb{R}^{d_{\text{a}}}$, and d_{a} is the alignment space dimension.

Cosine similarity: $\text{sim}(\mathbf{g}_i, \mathbf{c}_j) = \frac{\mathbf{g}_i^\top \mathbf{c}_j}{\|\mathbf{g}_i\|_2 \|\mathbf{c}_j\|_2}$

Let $\mathcal{B} \subseteq [N]$ be the set of node indices in the mini-batch; the symmetric contrastive loss is:

$$\mathcal{L}_{\text{align}} = \frac{1}{2}\left(\mathcal{L}_{\text{align}}^{g\to c} + \mathcal{L}_{\text{align}}^{c\to g}\right)$$

where:

$$\mathcal{L}_{\text{align}}^{g\to c} = -\frac{1}{|\mathcal{B}|} \sum_{i\in\mathcal{B}} \log \frac{\exp(\text{sim}(\mathbf{g}_i, \mathbf{c}_i)/\tau)}{\sum_{j\in\mathcal{B}} \exp(\text{sim}(\mathbf{g}_i, \mathbf{c}_j)/\tau)}$$

$\mathcal{L}_{\text{align}}^{c\to g}$ is defined symmetrically. $\tau > 0$ is the temperature hyperparameter.

Definition 6.10 (Total Training Loss).

$$\mathcal{L} = \mathcal{L}_{\text{cont}} + \lambda\, \mathcal{L}_{\text{disc}} + \mu\, \mathcal{L}_{\text{align}}$$

where $\lambda, \mu > 0$ are loss weight hyperparameters.

6.5 Training and Inference Algorithms

Algorithm 1: RiemannFM Single Training Step

Input: Training subgraph $(\mathbf{X}_1, \mathbf{E}_1, \mathbf{C}_\mathcal{V}, \mathbf{C}_\mathcal{R}, \mathbf{m})$, parameters θ **Output**: Loss $\mathcal{L}$

1. Sample $t \sim \text{Uniform}(0, 1 - \epsilon_t)$
2. For $i \in [N]$: sample $\mathbf{x}_{0,i} \sim p_0^\mathcal{M}$ (Definition 6.1)
3. For $(i, j) \in [N]^2$, $k \in [K]$: sample $\mathbf{E}_{0,ij}^{(k)} \sim \text{Bernoulli}(\rho_k)$ (Definition 6.2)
4. For $i \in [N]$: $\mathbf{x}_{t,i} = \exp_{\mathbf{x}_{0,i}}(t \cdot \log_{\mathbf{x}_{0,i}}(\mathbf{x}_{1,i}))$ (Definition 6.3)
5. For $(i, j) \in [N]^2$: sample $z_{ij} \sim \text{Bernoulli}(t)$, $\mathbf{E}_{t,ij} = z_{ij}\mathbf{E}_{1,ij} + (1 - z_{ij})\mathbf{E}_{0,ij}$ (Definition 6.4)
6. $(\hat{\mathbf{V}}, \hat{\mathbf{P}}) = f_\theta(\mathbf{X}_t, \mathbf{E}_t, t, \mathbf{C}_\mathcal{V}, \mathbf{C}_\mathcal{R}, \mathbf{m})$ (Definition 5.1)
7. For $i \in [N]$: $\mathbf{u}_{t,i} = \frac{1}{1-t} \log_{\mathbf{x}_{t,i}}(\mathbf{x}_{1,i})$ (Definition 6.5)
8. Compute $\mathcal{L} = \mathcal{L}_{\text{cont}} + \lambda\, \mathcal{L}_{\text{disc}} + \mu\, \mathcal{L}_{\text{align}}$ (Definition 6.10)
9. Riemannian Adam update for θ: standard Adam for Euclidean parameters, Riemannian gradient for curvature parameters
10. Curvature projection: $\kappa_h \leftarrow \min(\kappa_h, -\epsilon_\kappa)$, $\kappa_s \leftarrow \max(\kappa_s, \epsilon_\kappa)$

where $\epsilon_t > 0$, $\epsilon_\kappa > 0$ are small constants.

Algorithm 2: RiemannFM Inference

Input: Text conditioning $\mathbf{C}_\mathcal{V}, \mathbf{C}_\mathcal{R}$, number of steps $T \in \mathbb{Z}_{>0}$, step size $\Delta t = 1/T$
Output: $(\mathbf{X}_1, \mathbf{E}_1, \mathbf{m}_{\text{pred}})$

1. Sample $\mathbf{X}_0, \mathbf{E}_0$ (Definitions 6.1, 6.2)
2. **for** $s = 0, 1, \ldots, T - 1$ **do**:
   - $t = s \cdot \Delta t$
   - $(\hat{\mathbf{V}}, \hat{\mathbf{P}}) = f_\theta(\mathbf{X}_t, \mathbf{E}_t, t, \mathbf{C}_\mathcal{V}, \mathbf{C}_\mathcal{R}, \mathbf{m})$
   - For $i \in [N]$: $\mathbf{x}_{t+\Delta t,i} = \exp_{\mathbf{x}_{t,i}}(\Delta t \cdot \hat{\mathbf{v}}_i)$
   - For $(i, j) \in [N]^2$, $k \in [K]$: with probability $p_{\text{flip}}(t, \Delta t) = \min\left(\frac{\Delta t}{1-t}, 1\right)$ update $\mathbf{E}_{t+\Delta t,ij}^{(k)} = \mathbb{1}[\hat{\mathbf{P}}_{ij}^{(k)} > 0.5]$, otherwise keep $\mathbf{E}_{t,ij}^{(k)}$ unchanged
3. Virtual node removal: $m_{\text{pred},i} = \mathbb{1}[d_\mathcal{M}(\mathbf{x}_{1,i}, \mathbf{x}_\varnothing) \geq \epsilon_{\text{null}}]$
4. Edge binarization: $\mathbf{E}_{1,ij}^{(k)} \leftarrow \mathbb{1}[\hat{\mathbf{P}}_{ij}^{(k)} > p_{\text{thresh}}]$

where $\epsilon_{\text{null}} > 0$ is the virtual node detection threshold and $p_{\text{thresh}} \in (0, 1)$ is the edge binarization threshold.

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7. Downstream Task Fine-Tuning

7.1 Unified Masking Framework

Definition 7.1 (Task Masks). - Node task mask $\mathbf{m}^{\text{task}} \in \{0, 1\}^N$: $m_i^{\text{task}} = 1$ indicates to-be-predicted, $m_i^{\text{task}} = 0$ indicates known condition. - Edge task mask $\mathbf{M}^{\text{task}} \in \{0, 1\}^{N \times N}$: $M_{ij}^{\text{task}} = 1$ indicates to-be-predicted, $M_{ij}^{\text{task}} = 0$ indicates known condition.

Definition 7.2 (Conditioned Noisy Input). To-be-predicted parts are interpolated normally per Section 6; known parts retain the ground-truth data-end values:

$$\mathbf{x}_{t,i}^{\text{cond}} = \begin{cases} \exp_{\mathbf{x}_{0,i}}(t \cdot \log_{\mathbf{x}_{0,i}}(\mathbf{x}_{1,i})) & m_i^{\text{task}} = 1 \\ \mathbf{x}_{1,i} & m_i^{\text{task}} = 0 \end{cases}$$

$$\mathbf{E}_{t,ij}^{\text{cond}} = \begin{cases} z_{ij} \cdot \mathbf{E}_{1,ij} + (1 - z_{ij}) \cdot \mathbf{E}_{0,ij} & M_{ij}^{\text{task}} = 1 \\ \mathbf{E}_{1,ij} & M_{ij}^{\text{task}} = 0 \end{cases}$$

Definition 7.3 (Conditioned Fine-Tuning Loss). The loss is computed only on the to-be-predicted parts:

$$\mathcal{L}_{\text{cont}}^{\text{task}} = \frac{1}{\sum_i m_i^{\text{task}} \cdot m_i} \sum_{i=1}^{N} m_i^{\text{task}} \cdot m_i \cdot \|\hat{\mathbf{v}}_i - \mathbf{u}_{t,i}\|^2_{T_{\mathbf{x}_{t,i}}\mathcal{M}}$$

$$\mathcal{L}_{\text{disc}}^{\text{task}} = \frac{1}{\sum_{i,j} M_{ij}^{\text{task}}} \sum_{(i,j)} M_{ij}^{\text{task}} \sum_{k=1}^{K} \text{BCE}_{\text{w}}(\hat{\mathbf{P}}_{ij}^{(k)}, \mathbf{E}_{1,ij}^{(k)})$$

Note that $\mathcal{L}_{\text{cont}}^{\text{task}}$ requires both $m_i^{\text{task}} = 1$ (to-be-predicted) and $m_i = 1$ (real node, not virtual).

7.2 Task 1: Knowledge Graph Completion

Masking strategy. Given a training subgraph, construct task masks as follows:

- Edge mask: For each existing edge (i, j) (i.e., $\mathbf{E}_{1,ij} \neq \mathbf{0}_K$), independently set $M_{ij}^{\text{task}} = 1$ with probability p_{mask}^E;
- Node mask: For each real node i ($m_i = 1$), independently set $m_i^{\text{task}} = 1$ with probability p_{mask}^V, simultaneously masking all associated edges: $\forall j \in [N]$, $M_{ij}^{\text{task}} = M_{ji}^{\text{task}} = 1$;
- Remaining positions: $m_i^{\text{task}} = 0$, $M_{ij}^{\text{task}} = 0$.

where $p_{\text{mask}}^E, p_{\text{mask}}^V \in (0, 1)$ are hyperparameters.

Fine-tuning loss.

$$\mathcal{L}_{\text{KGC}} = \mathcal{L}_{\text{cont}}^{\text{task}} + \lambda_{\text{KGC}} \, \mathcal{L}_{\text{disc}}^{\text{task}}$$

where $\lambda_{\text{KGC}} > 0$ is a weight hyperparameter.

Evaluation. For a query fact $(v_i, r_k, ?)$, rank all candidate entities v_j by the model output $\hat{\mathbf{P}}_{ij}^{(k)}$. Metrics: - MRR $= \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \frac{1}{\text{rank}_q}$, where $\mathcal{Q}$ is the test query set; - Hits@$n = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \mathbb{1}[\text{rank}_q \leq n]$, $n \in \{1, 3, 10\}$.

7.3 Task 2: Text-Conditioned Subgraph Generation

Problem setting. Given a text query $q \in \Sigma^*$, generate a subgraph $\mathcal{G}$ that semantically matches the query.

Masking strategy. Full masking:

$$m_i^{\text{task}} = 1 \quad \forall i \in [N], \qquad M_{ij}^{\text{task}} = 1 \quad \forall (i,j) \in [N]^2$$

Text conditioning replacement. Replace all node text conditioning with the query embedding:

$$\mathbf{c}_q = \phi_{\text{text}}(q) \in \mathbb{R}^{d_c}, \qquad \mathbf{C}_{\mathcal{V}} = \mathbf{1}_N \mathbf{c}_q^\top \in \mathbb{R}^{N \times d_c}$$

where $\mathbf{1}_N \in \mathbb{R}^N$ is the all-ones vector. $\mathbf{C}_{\mathcal{R}}$ remains unchanged.

Fine-tuning loss. Since all parts are masked, the conditioned loss degenerates to the pretraining loss form:

$$\mathcal{L}_{\text{T2G}} = \mathcal{L}_{\text{cont}} + \lambda_{\text{T2G}}\, \mathcal{L}_{\text{disc}} + \mu_{\text{T2G}}\, \mathcal{L}_{\text{align}}$$

where $\lambda_{\text{T2G}}, \mu_{\text{T2G}} > 0$ are weight hyperparameters. The contrastive loss $\mathcal{L}_{\text{align}}$ encourages the generated node coordinates to be close to the query text $\mathbf{c}_q$ in the alignment space.

Evaluation. Align the generated subgraph with the ground-truth subgraph via maximum matching, then compute: - **Node F1**: Based on matched node text similarity, using a BERTScore threshold to determine successful matches; - **Relation F1**: On matched node pairs, compare the predicted edge type set with the ground-truth edge type set using F1; - **BERTScore**: Average BERTScore between node text descriptions in the generated and ground-truth subgraphs.

7.4 Task 3: Graph Anomaly Detection

Problem setting. Given a subgraph $\mathcal{G}$, detect anomalous edges and anomalous nodes. This task directly uses the pretrained model without fine-tuning.

Anomaly score construction. For the subgraph to be inspected, set $\mathbf{X}_1 = \mathbf{X}$, $\mathbf{E}_1 = \mathbf{E}$ as the data end, sample noise $\mathbf{X}_0, \mathbf{E}_0$ per Definitions 6.1–6.2, and construct noisy inputs $\mathbf{X}_t, \mathbf{E}_t$ per Definitions 6.3–6.4.

Definition 7.4 (Edge Anomaly Score). Let $\mathcal{T} = \{t_1, \ldots, t_{|\mathcal{T}|}\} \subset (0,1)$ be a preselected set of time steps (e.g., a uniform grid). For each existing edge (i,j,k) ($\mathbf{E}_{1,ij}^{(k)} = 1$):

$$S_{ij}^{(k)} = 1 - \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \hat{\mathbf{P}}_{ij}^{(k)}(f_\theta(\mathbf{X}_t, \mathbf{E}_t, t, \mathbf{C}_{\mathcal{V}}, \mathbf{C}_{\mathcal{R}}, \mathbf{m}))$$

Intuition: The pretrained model assigns high probability $\hat{\mathbf{P}}_{ij}^{(k)} \approx 1$ to normal facts, yielding anomaly score $S_{ij}^{(k)} \approx 0$; for anomalous facts, it assigns low

probability, yielding anomaly scores close to 1. Averaging over multiple time steps improves robustness.

Definition 7.5 (Node Anomaly Score). Let $\mathcal{N}_i = \{j \in [N] : \mathbf{E}_{1,ij} \neq \mathbf{0}_K\}$ be the set of outgoing-edge neighbors of node i.

$$S_i = \frac{1}{|\mathcal{N}_i|} \sum_{j \in \mathcal{N}_i} \max_{k:\, \mathbf{E}^{(k)}_{1,ij}=1} S^{(k)}_{ij}$$

That is, a node's anomaly level is the average anomaly score of the most suspicious relation among all its outgoing edges.

Evaluation. AUROC, AP (Average Precision).

Symbol Quick Reference

Base Sets and Indices

Symbol	Type	Domain	Meaning
$\mathcal{V}$	Set	—	Entity set
$\mathcal{R}$	Set	—	Relation type set
K	Scalar	$\mathbb{Z}_{>0}$	Total number of relation types, $K \triangleq \lvert\mathcal{R}\rvert$
$\mathcal{K}$	Graph	—	Knowledge graph $(\mathcal{V}, \mathcal{E})$
$\mathcal{E}$	Set	$\subseteq \mathcal{V} \times \mathcal{R} \times \mathcal{V}$	Fact set
$\mathcal{G}$	Graph	—	Subgraph $(\mathcal{V}_\mathcal{G}, \mathcal{E}_\mathcal{G})$
N	Scalar	$\mathbb{Z}_{>0}$	Subgraph node count (after padding, $N = N_{\max}$)
$N_{\max}$	Scalar	$\mathbb{Z}_{>0}$	Upper bound on subgraph node count
$[n]$	Set	—	$\{1, 2, \ldots, n\}$
$\mathcal{D}$	Set	—	Training dataset (collection of subgraphs)

Manifolds and Geometry

Symbol	Type	Domain	Meaning
$\mathcal{M}$	Manifold	—	Product manifold $\mathbb{H}^{d_h}_{\kappa_h} \times \mathbb{S}^{d_s}_{\kappa_s} \times \mathbb{R}^{d_e}$
d_h, d_s, d_e	Scalar	$\mathbb{Z}_{>0}$	Intrinsic dimensions of hyperbolic, spherical, Euclidean subspaces
D	Scalar	$\mathbb{Z}_{>0}$	Total ambient dimension, $D = (d_h+1)+(d_s+1)+d_e$
κ_h	Scalar	$\mathbb{R}_{<0}$	Hyperbolic space curvature (learnable)
κ_s	Scalar	$\mathbb{R}_{>0}$	Spherical curvature (learnable)
$\langle \cdot, \cdot \rangle_{\mathrm{L}}$	Operation	$\mathbb{R}$	Lorentz inner product
$T_{\mathbf{x}}\mathcal{M}$	Space	—	Tangent space at $\mathbf{x}$
$\exp_{\mathbf{x}}$	Map	$T_{\mathbf{x}}\mathcal{M} \to \mathcal{M}$	Exponential map
$\log_{\mathbf{x}}$	Map	$\mathcal{M} \to T_{\mathbf{x}}\mathcal{M}$	Logarithmic map
$d_{\mathcal{M}}$	Function	$\mathcal{M} \times \mathcal{M} \to \mathbb{R}_{\geq 0}$	Product manifold geodesic distance
$d_{\mathbb{H}}, d_{\mathbb{S}}, d_{\mathbb{R}}$	Function	—	Component subspace geodesic distances
π	Map	$\mathcal{M} \to \mathbb{R}^D$	Coordinate projection
$\mathbf{x}_{\varnothing}$	Vector	$\mathcal{M}$	Virtual node anchor

Subgraph Representation

Symbol	Type	Domain	Meaning
$\mathbf{x}_i$	Vector	$\mathcal{M}$	Manifold coordinate of node i
$\mathbf{x}_i^{\mathbb{H}}, \mathbf{x}_i^{\mathbb{S}}, \mathbf{x}_i^{\mathbb{R}}$	Vector	$\mathbb{R}^{d_h+1}, \mathbb{R}^{d_s+1}, \mathbb{R}^{d_e}$	Subspace components
$\mathbf{X}$	Tuple	$\mathcal{M}^N$	Node coordinate sequence
$\mathbf{E}$	Tensor	$\{0,1\}^{N \times N \times K}$	Edge type tensor
$\mathbf{E}_{ij}$	Vector	$\{0,1\}^K$	Multi-hot edge type from node i to j
$\mathbf{C}_{\mathcal{V}}$	Matrix	$\mathbb{R}^{N \times d_c}$	Node text conditioning matrix

Symbol	Type	Domain	Meaning
$\mathbf{C}_{\mathcal{R}}$	Matrix	$\mathbb{R}^{K \times d_c}$	Relation text conditioning matrix
$\mathbf{c}_i$	Vector	$\mathbb{R}^{d_c}$	Text conditioning of node i ($\mathbf{c}_i \triangleq \mathbf{c}_{v_i}$)
$\mathbf{c}_{r_k}$	Vector	$\mathbb{R}^{d_c}$	Text conditioning of relation r_k
$\mathbf{m}$	Vector	$\{0,1\}^N$	Virtual node mask
ϕ_{text}	Map	$\Sigma^* \to \mathbb{R}^{d_c}$	Pretrained text encoder
d_c	Scalar	$\mathbb{Z}_{>0}$	Text encoder output dimension

Model Architecture (RieFormer)

Symbol	Type	Domain	Meaning
f_θ	Map	—	RieFormer model
θ	—	—	All learnable parameters (including κ_h, κ_s)
L	Scalar	$\mathbb{Z}_{>0}$	Number of RieFormer layers
d_v	Scalar	$\mathbb{Z}_{>0}$	Node hidden dimension
$d_{e'}$	Scalar	$\mathbb{Z}_{>0}$	Edge hidden dimension
d_r	Scalar	$\mathbb{Z}_{>0}$	Relation embedding dimension
d_t	Scalar	$\mathbb{Z}_{>0}$ (even)	Time embedding dimension
d_{head}	Scalar	$\mathbb{Z}_{>0}$ (even)	Attention head dimension, $d_{\text{head}} = d_v/n_h$
d_{a}	Scalar	$\mathbb{Z}_{>0}$	Alignment space dimension
n_h	Scalar	$\mathbb{Z}_{>0}$	Number of attention heads
$\mathbf{h}_i^{V,(l)}$	Vector	$\mathbb{R}^{d_v}$	Layer l embedding of node i
$\mathbf{h}_{ij}^{E,(l)}$	Vector	$\mathbb{R}^{d_{e'}}$	Layer l embedding of edge (i,j)
$\mathbf{t}_{\text{emb}}$	Vector	$\mathbb{R}^{d_v}$	Time embedding

Symbol	Type	Domain	Meaning
$\hat{\mathbf{v}}_i$	Vector	$T_{\mathbf{x}_{t,i}}\mathcal{M}$	Model-predicted tangent vector
$\hat{\mathbf{P}}_{ij}^{(k)}$	Scalar	$[0,1]$	Edge type probability prediction
$\kappa^{(s)}$	Function	$\mathcal{M} \times \mathcal{M} \to \mathbb{R}$	Geodesic kernel for head s
$\hat{\rho}_i$	Scalar	$\mathbb{R}_{\geq 0}$	Hierarchical depth estimate
$\odot$	Operation	—	Element-wise product (Hadamard product)
$\|$	Operation	—	Vector concatenation

Flow Matching

Symbol	Type	Domain	Meaning
t	Scalar	$[0,1]$	Flow matching time step (0=noise, 1=data)
$\mathbf{X}_0, \mathbf{E}_0$	—	—	Noise-end samples
$\mathbf{X}_1, \mathbf{E}_1$	—	—	Data-end (training subgraph)
$\mathbf{x}_{t,i}$	Vector	$\mathcal{M}$	Interpolated coordinate of node i at time t
$\mathbf{E}_{t,ij}$	Vector	$\{0,1\}^K$	Interpolated edge type of (i,j) at time t
z_{ij}	Scalar	$\{0,1\}$	Edge interpolation mask, $z_{ij} \sim \text{Bernoulli}(t)$
$\mathbf{u}_{t,i}$	Vector	$T_{\mathbf{x}_{t,i}}\mathcal{M}$	Conditional vector field target
ρ_k	Scalar	$[0,1]$	Marginal frequency of relation r_k
$p_0^{\mathcal{M}}$	Distribution	—	Continuous noise prior
$R_{\mathbb{H}}$	Scalar	$\mathbb{R}_{>0}$	Hyperbolic uniform distribution truncation radius

Symbol	Type	Domain	Meaning
σ_0	Scalar	$\mathbb{R}_{>0}$	Euclidean noise standard deviation
T	Scalar	$\mathbb{Z}_{>0}$	Number of inference steps
Δt	Scalar	$(0, 1]$	Inference step size, $\Delta t = 1/T$
p_{flip}	Function	$[0, 1]$	Edge update flip probability

Loss Functions

Symbol	Type	Domain	Meaning
$\mathcal{L}_{\text{cont}}$	Scalar	$\mathbb{R}_{\geq 0}$	Manifold vector field loss
$\mathcal{L}_{\text{disc}}$	Scalar	$\mathbb{R}_{\geq 0}$	Edge type loss
$\mathcal{L}_{\text{align}}$	Scalar	$\mathbb{R}_{\geq 0}$	Graph-text contrastive loss
$\mathcal{L}$	Scalar	$\mathbb{R}_{\geq 0}$	Total training loss
λ, μ	Scalar	$\mathbb{R}_{>0}$	Loss weights
τ	Scalar	$\mathbb{R}_{>0}$	Contrastive loss temperature
w_k^+	Scalar	$\mathbb{R}_{>0}$	Positive sample weight for relation r_k
$w_{\max}$	Scalar	$\mathbb{R}_{>0}$	Positive sample weight truncation upper bound
$\mathcal{S}$	Set	$\subseteq [N]^2$	Edge loss sampling set
$\mathcal{B}$	Set	$\subseteq [N]$	Contrastive loss mini-batch indices
$\epsilon_t, \epsilon_p, \epsilon_\kappa$	Scalar	$\mathbb{R}_{>0}$	Numerical stability/truncation constants

Downstream Tasks

Symbol	Type	Domain	Meaning
$\mathbf{m}^{\text{task}}$	Vector	$\{0, 1\}^N$	Node task mask
$\mathbf{M}^{\text{task}}$	Matrix	$\{0, 1\}^{N \times N}$	Edge task mask

Symbol	Type	Domain	Meaning
$\mathcal{L}_{\text{cont}}^{\text{task}}, \mathcal{L}_{\text{disc}}^{\text{task}}$	Scalar	$\mathbb{R}_{\geq 0}$	Conditioned fine-tuning loss
$p_{\text{mask}}^{E}, p_{\text{mask}}^{V}$	Scalar	$(0, 1)$	KGC masking probabilities
$\mathbf{c}_q$	Vector	$\mathbb{R}^{d_c}$	T2G query text embedding
$S_{ij}^{(k)}$	Scalar	$[0, 1]$	Edge anomaly score
S_i	Scalar	$[0, 1]$	Node anomaly score
$\mathcal{T}$	Set	$\subset (0, 1)$	GAD preselected time step set
$\mathcal{N}_i$	Set	$\subseteq [N]$	Outgoing-edge neighbor set of node i
ϵ_{null}	Scalar	$\mathbb{R}_{>0}$	Virtual node detection threshold
p_{thresh}	Scalar	$(0, 1)$	Edge binarization threshold